Project Proposal

MediBridge:
An AI Healthcare Navigator

MediBridge is a digital health navigator that helps international students find appropriate medical clinics and estimate out-of-pocket costs through a simple. By clarifying insurance coverage and billing options, the platform empowers students to access the Australian healthcare system with confidence and financial certainty.

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1. Introduction

1.1. Industry Context & Background

It is a well-known fact that the Australian healthcare system is thoroughly organised with structures like the Medicare Benefits Schedule (MBS) and Overseas Student Health Cover (OSHC) governing the healthcare system. Although this system has a wide range of healthcare provision, international students find it difficult to access since it is complicated and foreign to them. Multinational students commonly face challenges in comprehending the healthcare pathways, insurance, and cost arrangements, leading to a decrease in the use of healthcare and a rise in the use of informal advice networks.

The empirical data reveal that out of the number of international students who strongly agree they have learned about the Australian healthcare system; the percentage is only approximately 17 per cent. In addition, around 82 per cent of them are concerned about healthcare cost-effectiveness, and about 29 per cent would prefer to access healthcare services in their countries of origin rather than through local healthcare systems (2025-State-of-Student-Healthcare-Report, 2025). The statistics provided indicate that there is a critical lack of health literacy and access, both being directly related to the well-being of the students, their performance in school, and their overall experiences.

Cultural and linguistic barriers are also another problem that aggravates the situation. The international students have medical systems that have very different structures, which creates confusion on how the appointments are made, the process of billing, and the reimbursement of the insurance (Men et al., 2024). This means that there is an urgent requirement for a digital solution that will close this gap in knowledge and offer culturally contextualised healthcare navigation.

1.2. Aim of the Project

MediBridge is an AI-powered health navigator designed to improve healthcare accessibility for international students in Australia by mitigating systemic confusion and financial anxiety. Through a digital interface, the system transforms complex insurance and medical information into actionable pre-visit guidance, helping users identify appropriate services and estimate potential costs. By focusing strictly on navigation and confidence-building, MediBridge assists in informed decision-making without offering clinical diagnostics, treatment plans, or medication recommendations.

1.3. Objectives

- To create a conversational chatbot based on AI and powered with RAG to provide precise and policy-centred healthcare advice.
- To introduce a Smart Clinic Locator where nearby GPs and specialists with bulk-billed and private-billed filters could be found.
- To prepare a Gap Cost Calculator to determine out-of-pocket costs with reference to MBS and OSHC covers.
- To deliver a Minimum Viable Product (MVP) by the end of the semester, demonstrating the core value of the solution through a fully functional web interface with at least basic guidance functionality through the AI Chat.
- To provide a scalable final project direction to support broader multilingual capabilities, more robust healthcare provider search functionality, and more comprehensive data-driven healthcare guidance through cost and savings calculation.

1.4. Potential Impacts and Outcomes

MediBridge delivers a practical digital solution that enhances healthcare accessibility for international students by alleviating financial anxiety and fostering informed decision-making. On the other hand, healthcare providers gain operational efficiency through patients who arrive with clearer insurance and billing expectations. By integrating AI to solve real-world navigation challenges, the project advances digital health innovation, promoting transparency and increased engagement across the healthcare ecosystem.

2. Project Management Approach

2.1. Selected Project Management Approach

The project utilizes Agile Scrum, a framework characterized by iterative development, time-boxed cycles (sprints), and a focus on incremental delivery. This approach allows the team to break down the complex MediBridge system—comprising a conversational chatbot, clinic locator, and cost estimator—into manageable, functional components.

2.2. Justification for Industry Fit

Scrum is the optimal fit for MediBridge due to several project-specific factors:

- **AI Uncertainty:** Developing a Retrieval-Augmented Generation (RAG) chatbot involves constant experimentation with prompt engineering and data retrieval. Scrum's iterative nature allows us to test and revamp these processes through trial and error.
- **Dynamic Data Sources:** Healthcare data, such as MBS (Medicare Benefits Schedule) datasets and OSHC (Overseas Student Health Cover) guidelines, are subject to periodic updates. Scrum's flexible product backlog ensures the system remains compliant and accurate.
- **Stakeholder Integration:** Frequent feedback from academic supervisors and industry mentors is vital. Scheduled sprint reviews ensure that the navigation system aligns with user needs and regulatory standards in real-time.

2.3. Adaptation and Customization

To ensure the successful delivery of MediBridge's unique requirements, the standard Scrum framework has been customized:

- **Enhanced Validation Sprints:** Beyond standard functional testing, we have integrated rigorous validation measures into every sprint to cross-reference AI outputs with official Australian healthcare guidelines and privacy standards.
- **Research-Focused Increments:** We have adopted a hybrid element where specific sprints are dedicated to R&D, such as analyzing new AI models or data integration strategies, allowing us to manage technical debt without halting overall progress.
- **Specialized "Definition of Done" (DoD):** Our DoD is expanded to include specific metrics for AI response accuracy and data privacy adherence, ensuring that every increment meets healthcare-grade safety levels.

2.4. PM Tools and Controls

To maintain transparency and track progress, the following tools and controls will be implemented:

- **Jira:** Used for managing the product backlog, planning two-week sprints, and visualizing task distribution via Kanban boards.
- **GitHub:** Facilitates version control and collaborative coding, utilizing pull requests and code reviews to maintain technical integrity.

3. Project Scope and Plan

The MediBridge project utilizes a structured Agile approach to define boundaries and deliverables, ensuring the AI health navigator remains focused on informational support rather than clinical intervention.

3.1. Scope Definition

To manage project complexity and regulatory risk, the scope is strictly categorized as follows:

In Scope	Out of Scope	Scope Risks & Unknowns
Conversational OSHC guidance.	Medical diagnosis or clinical advice.	Depth of multilingual support.
GP/Specialist search with billing filters.	Medication/prescription recommendations.	Reliability of 3rd-party clinic data.
MBS-based cost estimation logic.	Direct appointment booking.	Real-time predictive data feasibility.
Privacy-first web interface.	Personal health record (PHR) management.	Validation of AI-generated accuracy.

3.2. System Requirements

3.2.1. Functional Requirements

- **FR1 – Conversational Health Guide:** The system must process natural language queries regarding OSHC and healthcare pathways, providing simple, non-diagnostic informational support.
- **FR2 – Smart Clinic Search:** Users must be able to search for nearby GPs and specialists using filters for billing types (e.g., bulk vs. private billing).
- **FR3 – Out-of-Pocket Expense Calculator:** The system must estimate potential "gap fees" by comparing clinic price data against Medicare Benefits Schedule (MBS) rebate logic.

3.2.2. Non-Functional Requirements (NFR)

These define the quality attributes and constraints of the system:

- **Accuracy:** All AI responses must be grounded in official Australian healthcare and insurance datasets to prevent "hallucinations."
- **Security & Privacy:** The platform must comply with **Australian Privacy Principles (APP)**, minimizing personal data collection and ensuring secure data handling.
- **Latency:** Core interactions (chat and search) must provide low-latency responses to maintain user trust and usability.

- **Scalability & Concurrency:** The architecture must remain stable under simultaneous access by multiple users, particularly during peak university orientation periods.

3.3. Project Plan: Sprint Schedule

The project is divided into seven two-week sprints to facilitate iterative improvement and stakeholder feedback.

SPRINT	FOCUS	PLANNED DELIVERABLES
SPRINT 1	Requirements & Research	Validated problem statement and finalized user stories.
SPRINT 2	UI & Architecture	Wireframes, system architecture, and tech stack setup.
SPRINT 3	Chat Module MVP	Initial chatbot interface and RAG grounding design.
SPRINT 4	Clinic Finder MVP	Searchable GP list with functional billing filters.
SPRINT 5	Cost Calculator MVP	Fee estimation logic and initial user testing.
SPRINT 6	Integration & QA	Integrated prototype and rigorous usability review.
SPRINT 7	Final Refinement	Final deployment demo and comprehensive project report.

3.4. Deliverables and Estimation

3.4.1. Interim Artefacts (MVP - Week 14)

- Responsive Web Interface: A functional front-end for user interaction.
- Core Navigation Engine: Baseline chat, provider search, and cost estimation modules.
- QA Documentation: Initial testing reports and deployment status.

3.4.2. Final Artefacts (End of Project)

- Advanced RAG Engine: A robust AI system with improved response grounding.
- Multilingual Support: Expanded UI for diverse international student groups.
- Technical Portfolio: Complete architecture diagrams, code documentation, and compliance audits.

3.4.3. Initial Effort Estimation

Requirement / Epic	Complexity	Estimated Effort
Health Navigation Chat	**High**	**4 Weeks**
GP & Specialist Finder	**Medium**	**2–3 Weeks**
Cost Calculator	**Medium**	**2 Weeks**
System Integration & QA	**High**	**2 Weeks**

4. Technical Solution

The MediBridge solution is a high-performance, **AI Health Navigator** designed to solve the specific accessibility challenges faced by international students in Australia. Unlike traditional chatbots that rely on static text, MediBridge uses a "Reasoning Agent" architecture that can interact with complex insurance datasets to provide actionable guidance. We also make use healthcare APIs to get access to clinic location. Using the healthcare datasets, we provide the users with a cost estimation calculator.

4.1. Proposed System Architecture

- Frontend: Built with React.js and Tailwind CSS. This stack provides a highly responsive, "App-like" web experience.
- Backend: A FastAPI (Python) server acts as the orchestrator, managing secure connections between the user, the AI agent, and external data sources.
- The "Brain" (Agentic RAG): We utilize LangGraph combined with GPT-5-mini. This allows the system to function as an "Agent" that can reason through a user's request, decide which tools to use, and maintain the "state" of the conversation (e.g., remembering a user's insurance provider while calculating costs).

4.2. Justification of Technology Choices

Our team conducted a critical analysis of current industry standards to select a stack that balances performance with clinical safety.

Component	Chosen Technology	Justification & Critical Analysis
LLM Engine	GPT-5-mini	Chosen for its advanced Reasoning Mode and native support for Function Calling. It can reliably translate natural language into structured API queries for clinic searches.
Orchestration	LangChain / LangGraph	Standard RAG is "linear" and often fails at complex tasks. LangGraph enables a "cyclic" workflow, allowing the agent to self-correct and verify information across multiple steps.
Frontend	React.js	Provides the best path for Future Scalability. Because React.js shares logic with React Native, we can transition the MVP into a full mobile app with minimal refactoring of the core logic.
Clinic Data	HealthDirect API	Instead of a manual database, we leverage the HealthDirect API (the gold standard for Australian provider data). This ensures clinic hours, locations, and billing types are always 100% accurate.
Knowledge Base	Vector Databases	Essential for indexing the Medicare Benefits Schedule (MBS) and OSHC policy manuals. It allows the agent to perform "Semantic Search" to find the correct billing codes for cost estimation.

4.3. Design Strategy

Our solution demonstrates knowledge from advanced IT units by implementing a Separation of Concerns. By using FastAPI as a middleware, we ensure that sensitive user queries are processed securely before reaching the LLM. Furthermore, by architecting the frontend in React.js, we have significantly reduced the "Technical Debt" for the project's next phase—the transition to a native mobile application. The modularity of our Agentic RAG means that as Australian healthcare policies change, we only need to update the documents in the Vector Database, ensuring MediBridge remains a reliable, long-term solution.

5. Communication and Risk Management

5.1. Stakeholder Communication Plan

Our communication plan addresses the "Who, What, Where, When, and How" of the project lifecycle. By integrating regular feedback loops, we ensure that Quality Assurance (QA) is embedded in every sprint, not just at the end of the project.

STAKEHOLDER	WHAT (PURPOSE)	WHERE (METHOD)	WHEN (FREQUENCY)
PROJECT TEAM	Task coordination, blocker resolution, and technical alignment.	WhatsApp, Discord, & Zoom	Daily (Async) / Weekly (Sync)
ACADEMIC SUPERVISOR (TUTOR)	Academic alignment, scope validation, and milestone grading.	Zoom or In-person	Weekly
INDUSTRY MENTOR / PARTNER	Requirement refinement, technical validation, and industry feedback.	Email & Scheduled Meetings	Fortnightly (or Weekly if required)
INTERNATIONAL STUDENTS (PRIMARY USERS)	Usability testing and feedback on clarity/navigation.	Surveys & Beta Testing	End of Sprints 4 & 6

- **Internal:** We use Discord for persistent technical documentation and WhatsApp for urgent "blocker" communication. Our twice-weekly meetings act as the anchor for our Scrum ceremonies (Planning and Retrospectives).
- **External:** Communication with supervisors and mentors is formal and focused on **documented progress**. Every feedback round results in a "backlog refinement," ensuring the product evolves according to professional expectations.

5.2. Risk Analysis and Mitigation

We categorize risks into three levels: Team, Individual, and Industry/Project. In consultation with our project supervisor, our "Line of Tolerance" is defined as any risk with a "High" impact or a "High" likelihood. All risks above this line require a dedicated mitigation strategy.

Risk	Likelihood	Impact	Mitigation Strategy
AI Hallucination	Medium	High	Use Agentic RAG (LangGraph) to ground responses in OSHC/MBS docs; enforce "no-medical-advice" system prompts.
Data Privacy Breach	Low	High	Stateless API design: do not store chat history or PII; implement end-to-end encryption for active sessions.
Timeline Slippage	High	Medium	Adopt Scrum Sprints with a 20% buffer; prioritize the Chat MVP over advanced search features for Week 14.
API Dependency	Medium	Medium	Implement mock data fallbacks for the HealthDirect API to ensure the system remains functional during external downtime.
Inaccurate Costing	Medium	High	Cross-verify Vector DB retrieval logic against manual MBS lookups during Sprint 5 QA; include "Estimated" disclaimers.
Scope Creep	Medium	Medium	Strict adherence to the MoSCoW method (Must-have vs. Should-have) during weekly Product Backlog refinement.

5.3. Risk Management Strategy

- **Avoidance:** For risks like Data Privacy, we avoid the risk entirely by choosing not to store any Personal Health Information (PHI) in our database.
- **Mitigation:** For AI Hallucination, we use "System Prompts" and "Guardrails" within LangGraph to ensure the AI explicitly states it is not a doctor and only cites official OSHC documents.
- **Transference:** For technical hosting risks, we use industry-standard platforms like AWS or Vercel, transferring the server uptime responsibility to the provider.
- **Acceptance:** We acknowledge that Timeline Slippage is a constant threat in Agile; we manage this by focusing on the Minimum Viable Product (MVP) first to ensure we have a deliverable by Week 14.

6. Cost Estimation

The following table outlines the resource allocation and operational overhead. These figures represent high-level estimates based on current industry benchmarks for digital health innovation projects in Australia.

Category	Description	Estimated Cost (AUD)
Development Cost	Conceptual modeling of full-stack, AI, and integration engineering efforts.	$13,500
AI & Data Services	Token consumption for GPT-5-mini and Vector Database hosting fees.	$300
Cloud Infrastructure	Scalable hosting (AWS/Vercel) and deployment pipeline maintenance.	$200
Operational Tools	Licenses for project management, version control, and UI frameworks.	$100
Contingency (10%)	Strategic buffer for technical pivots, R&D, and unforeseen risks.	$1,400
TOTAL ESTIMATED VALUE		**$15,500**

7. Appendix

Appendix A: Team Contract

8. References

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